

# DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

## Supplementary Summer Examination – 2025

Course: B. Tech

Branch: Civil Engineering

Semester: VI

Subject Code & Name: BTCVE605A\_Y18 & Waste Water Treatment

Max Marks: 60

Date: 10/07/25

Duration: 3 Hr.

### Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|                                                                 | (Level /CO) | Marks     |
|-----------------------------------------------------------------|-------------|-----------|
| <b>Q. 1 Objective type questions. (Compulsory Question)</b>     |             | <b>12</b> |
| 1 Wastewater treatment mainly aims to:                          | L1          | 1         |
| a. Increase water consumption                                   |             |           |
| b. Remove harmful contaminants                                  |             |           |
| c. Produce drinking water only                                  |             |           |
| d. Add nutrients to water                                       |             |           |
| 2 Septic tank is mainly used for:                               | L1          | 1         |
| a. Final treatment of water                                     |             |           |
| b. Preliminary treatment only                                   |             |           |
| c. Treatment in rural areas                                     |             |           |
| d. Removal of heavy metals                                      |             |           |
| 3 Oxidation ditch is an example of:                             | L1          | 1         |
| a. Primary treatment                                            |             |           |
| b. Tertiary treatment                                           |             |           |
| c. Low cost wastewater treatment                                |             |           |
| d. Membrane process                                             |             |           |
| 4 The DO sag curve is related to:                               | L1          | 1         |
| a. Storm water drainage                                         |             |           |
| b. Stream pollution and self-purification                       |             |           |
| c. Soil contamination                                           |             |           |
| d. Air pollution control                                        |             |           |
| 5 Neutralization in industrial wastewater treatment is used to: | L1          | 1         |
| a. Remove oil                                                   |             |           |
| b. Adjust pH                                                    |             |           |
| c. Remove solids                                                |             |           |
| d. Increase BOD                                                 |             |           |
| 6 Phosphorus removal is generally part of:                      | L1          | 1         |
| a. Preliminary treatment                                        |             |           |
| b. Secondary treatment                                          |             |           |
| c. Advanced treatment                                           |             |           |
| d. Screening                                                    |             |           |
| 7 CETPs stand for:                                              | L1          | 1         |
| a. Community Effluent Testing Plants                            |             |           |
| b. Common Effluent Treatment Plants                             |             |           |
| c. Combined Emission Testing Points                             |             |           |
| d. Central Effluent Tertiary Plants                             |             |           |
| 8 Mosquitoes mainly spread:                                     | L1          | 1         |
| a. Typhoid                                                      |             |           |
| b. Malaria                                                      |             |           |
| c. Ringworm                                                     |             |           |
| d. Tuberculosis                                                 |             |           |

|             |                                                                                               |                                          |           |
|-------------|-----------------------------------------------------------------------------------------------|------------------------------------------|-----------|
| 9           | Oxidation pond mainly works on:                                                               | L1                                       | 1         |
|             | a. Membrane filtration                                                                        | b. Anaerobic digestion                   |           |
|             | c. Natural biological processes                                                               | d. Chemical oxidation                    |           |
| 10          | Low cost excreta disposal systems are especially needed in:                                   | L1                                       | 1         |
|             | a. Urban areas                                                                                | b. Coastal cities                        |           |
|             | c. Rural areas                                                                                | d. Industrial zones                      |           |
| 11          | The main difference between sewage and sullage is:                                            | L1                                       | 1         |
|             | a. Sullage contains storm water                                                               | b. Sewage does not contain human excreta |           |
|             | c. Sullage does not contain human excreta                                                     | d. Both are same                         |           |
| 12          | Ion exchange method is mainly used for removing:                                              | L1                                       | 1         |
|             | a. Nutrients                                                                                  | b. Pathogens                             |           |
|             | c. Heavy metals                                                                               | d. Floating debris                       |           |
| <b>Q. 2</b> | <b>Solve the following.</b>                                                                   |                                          | <b>12</b> |
| A)          | Write note on various physical properties of waste water.                                     | L1                                       | 6         |
| B)          | Discuss various preliminary treatment of wastewater.                                          | L2                                       | 6         |
| <b>Q.3</b>  | <b>Solve the following.</b>                                                                   |                                          | <b>12</b> |
| A)          | Explain the principle of stabilization pond.                                                  | L2                                       | 6         |
| B)          | Explain briefly Oxidation Pond with following points:                                         | L3                                       | 6         |
|             | a) Working Operation b) Merits and Demerits.                                                  |                                          |           |
| <b>Q. 4</b> | <b>Solve Any Two of the following.</b>                                                        |                                          | <b>12</b> |
| A)          | Discuss in brief the biological & chemical method of removal of phosphorous from waste water. | L1                                       | 6         |
| B)          | Write a short note on membrane separation process.                                            | L2                                       | 6         |
| C)          | What do you mean by common effluent treatment plant? What is the need of CETP?                | L3                                       | 6         |
| <b>Q.5</b>  | <b>Solve Any Two of the following.</b>                                                        |                                          | <b>12</b> |
| A)          | Discuss communicable diseases and methods of communication.                                   | L1                                       | 6         |
| B)          | Explain methods of fly control and engineering approaches to prevent fly breeding.            | L2                                       | 6         |
| C)          | Describe plague control methods and role of rodents in public health.                         | L3                                       | 6         |
| <b>Q. 6</b> | <b>Solve Any Two of the following.</b>                                                        |                                          | <b>12</b> |
| A)          | Explain in brief essentials of rural sanitation.                                              | L1                                       | 6         |
| B)          | Outline main features & provisions under swachh Bharat Abhiyan (Clean India Mission).         | L2                                       | 6         |
| C)          | Discuss low cost excreta disposal systems suitable for rural areas.                           | L3                                       | 6         |

\*\*\* End \*\*\*